PROBLEM DA14

Torque in an Impulsively Rotated Cavity

Original statement · DA14.pdf

Π theorem | Navier-Stokes | Nondimensionalization | Viscous dissipation

OFFICIAL PROBLEM STATEMENT

The sketch shows a circular cavity full of liquid of density ρ and viscosity μ surrounded by an outer case at temperature T_0. The fluid is initially at rest, temperature T_0 and pressure p_0. At $t = 0^+$, the outer case is impulsively set to rotate with a constant angular velocity Ω.

1. Use the Π theorem to reduce the parametric dependence of the torque M (per unit length) necessary to sustain the constant angular velocity of the outer case from $t = 0$ to $t \to \infty$.

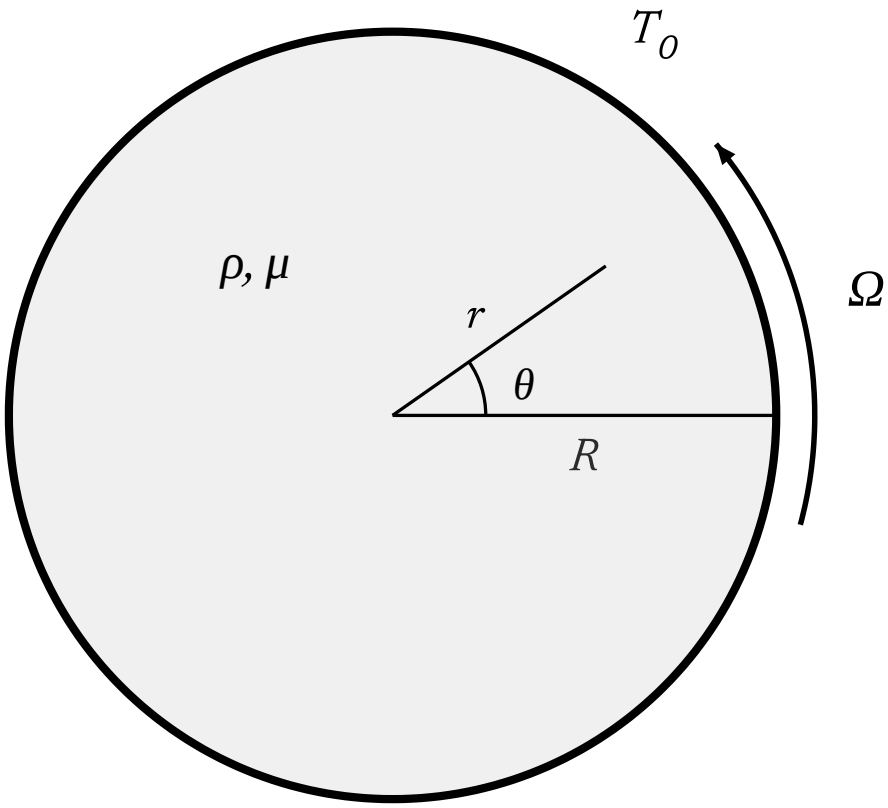
2. Formulate the **full** mathematical problem that determines the evolution of the fluid field (velocity, pressure and temperature) in the cavity $0 \leq r \leq R$ for $0 < t < \infty$, including boundary and initial conditions. Consider the pressure in the fluid adjacent to the wall is known to be p_1. Simplify the problem as much as possible to obtain the differential equations, with initial and boundary conditions, that determine the azimuthal velocity v_θ, T and p. Viscous dissipation might not be negligible.

3. Find v_θ, T and p in the case $t \to \infty$.

4. From the last result, obtain the adequate scales $v_{\theta c}$, r_c, respectively for v_θ, r. Find the adequate time scale t_c using order-of-magnitude analysis in the time-dependent differential equation for v_θ. Make the problem non-dimensional introducing the scales in the dimensional problem.

5. Define the expression that needs to be evaluated to find the torque M at any instant of time t, using the detailed solution of the non-dimensional problem, scaling M adequately.

6. Compare the last result with that of part #1 and comment the result



Statement figure (redrawn from DA14.pdf): a circular cavity of radius R full of liquid (ρ, μ), initially at rest, inside a case at temperature T_0 that is set to rotate at Ω at t = 0⁺

Note: It is strongly recommended to use cylindrical coordinates

`Source: DA14.pdf (page 1: statement; pages 2 and 3: handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; nondimensionalization, Eqs. 6.45-6.50. Notes.pdf, Chapter 4: continuity, momentum and energy equations of a perfect liquid.`

Worked solution

Hypotheses, Data and Dimensional Table

- Perfect liquid: constant ρ, μ, thermal conductivity k and specific heat c. With constant properties the temperature does not act back on the velocity, so the torque does not depend on T_0, k or c.
- The cavity is long (the torque is per unit length) and the motion is axisymmetric: $\partial/\partial\theta = 0, v_z = 0, \partial/\partial z = 0$. Gravity is not considered (it would only add a hydrostatic part to p).
- Frame: cylindrical coordinates (r, θ, z) fixed to the ground, origin on the axis.

Quantity	**Dimensions**
M (torque per unit length)	MLT^{-2}
t	T
ρ	ML^{-3}
μ	$ML^{-1}T^{-1}$
Ω	T^{-1}
R	L

Application of the Π Theorem

$$M = M(t, \rho, \mu, \Omega, R)$$

The torque per unit length has the dimensions of a force: $[M] = ML^2T^{-2}/L = MLT^{-2}$. Exponent matrix with columns (M, L, T), as in the key:

	M	L	T
M	1	1	-2
t	0	0	1
ρ	1	-3	0
μ	1	-1	-1

	M	L	T
Ω	0	0	-1
R	0	1	0

The rows of (μ, Ω, R) give, expanding along the first column, $\det = 1 \cdot (0 \cdot 0 - (-1) \cdot 1) = 1 \neq 0$: they have independent dimensions. With $n = 5$ and $k = 3$, two groups remain.

Torque. $[M] = [\mu]^a [\Omega]^b [R]^c$: $M:\ 1 = a$; $T:\ -2 = -a - b, b = 1$; $L:\ 1 = -a + c, c = 2$: $\Pi_0 = M/(\mu \Omega R^2)$.

Time. $[t] = [\Omega]^{-1}$: $\Pi_1 = \Omega t$.

Density. $[\rho] = [\mu]^a [\Omega]^b [R]^c$: $M:\ 1 = a$; $T:\ 0 = -1 - b, b = -1$; $L:\ -3 = -1 + c$, $c = -2$: $\Pi_2 = \rho \Omega R^2/\mu$, a Reynolds number.

$$\frac{M}{\mu \Omega R^2} = f\left(\Omega t, \frac{\rho \Omega R^2}{\mu}\right)$$

The Full Problem and its Simplification

The equations of a perfect liquid (Notes.pdf, Chapter 4) in cylindrical coordinates, with $\partial/\partial\theta = 0$ and $v_z = 0$:

Continuity.

$$\frac{1}{r}\frac{\partial (r v_r)}{\partial r} = 0 \implies r v_r = f(t)$$

At the wall $v_r(R, t) = 0$, so $f(t) = 0$ and $v_r = 0$ everywhere.

Radial momentum. With $v_r = 0$ the only surviving inertial term is the centripetal one:

$$-\rho \frac{v_\theta^2}{r} = -\frac{\partial p}{\partial r}$$

Azimuthal momentum. The convective terms $v_r \partial v_\theta/\partial r$, $v_r v_\theta/r$ and the pressure term $(1/r)\partial p/\partial \theta$ vanish:

$$\frac{\partial v_\theta}{\partial t} = \nu \frac{\partial}{\partial r}\left[\frac{1}{r}\frac{\partial (r v_\theta)}{\partial r}\right]$$

Energy. The convective terms vanish ($v_r = 0$, $\partial T/\partial \theta = 0$); conduction and viscous dissipation remain:

$$\rho c \frac{\partial T}{\partial t} = \frac{k}{r}\frac{\partial}{\partial r}\left(r \frac{\partial T}{\partial r}\right) + \mu \left[r \frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)\right]^2$$

The last term is the dissipation Φ_v of the only non-zero strain rate, $r\,\partial(v_\theta/r)/\partial r$.

Initial and boundary conditions.

Where	Condition	Origin
$t = 0$	$v_\theta = 0, T = T_0$	fluid at rest at T_0
$r = R, t > 0$	$v_\theta = \Omega R, T = T_0$	no slip on the rotating case; case at T_0
$r = R$	$p = p_1$	statement
$r = 0$	$v_\theta = 0, \partial T/\partial r = 0$	regularity on the axis

The velocity problem is closed by itself; p follows from the radial equation and T from the energy equation once v_θ is known.

The Limit t → ∞

Velocity. Steady state: $\partial/\partial t = 0$ in the azimuthal equation, so

$$\frac{\partial}{\partial r}\left[\frac{1}{r}\frac{\partial (r v_\theta)}{\partial r}\right] = 0 \implies \frac{1}{r}\frac{d(r v_\theta)}{dr} = A$$

Multiply by r and integrate from the axis, where $r v_\theta = 0$, to r (Barrow's rule):

$$\int_0^r \frac{d(r' v_\theta)}{dr'}\, dr' = \left[r' v_\theta\right]_0^r = r v_\theta - 0 = \int_0^r A r'\, dr' = \left[\frac{A r'^2}{2}\right]_0^r = \frac{A r^2}{2}$$

so $v_\theta = Ar/2$; the general solution $Ar/2 + B/r$ has lost its B/r term, singular on the axis. The wall condition $v_\theta(R) = \Omega R$ gives $A = 2\Omega$:

$$v_\theta = \Omega r$$

the liquid rotates as a rigid body.

Pressure. $\partial p/\partial r = \rho v_\theta^2/r = \rho \Omega^2 r$. Integrating from the wall, where $p = p_1$, to r:

$$p(r) - p_1 = \int_R^r \rho \Omega^2 r'\, dr' = \left[\frac{\rho \Omega^2 r'^2}{2}\right]_R^r = \frac{\rho \Omega^2}{2}\left(r^2 - R^2\right) \implies p = p_1 + \frac{1}{2}\rho \Omega^2 R^2 \left[\left(\frac{r}{R}\right)^2 - 1\right]$$

Temperature. In rigid rotation $v_\theta/r = \Omega$ is uniform, so $\Phi_v = 0$, and the steady energy equation is $\frac{d}{dr}\left(r \frac{dT}{dr}\right) = 0$. Integrating from the axis, where regularity gives $r\, dT/dr = 0$, to r (Barrow's rule): $\left[r' \frac{dT}{dr'}\right]_0^r = r \frac{dT}{dr} - 0 = 0$, so $dT/dr = 0$ for $r > 0$, T is uniform, and the wall condition gives

$$T = T_0$$

Scales and Nondimensional Problem

From the steady solution: $r_c = R$ and $v_{\theta c} = \Omega R$. The time scale comes from balancing the two terms of the azimuthal equation (Notes.pdf, Eq. 6.45):

$$\frac{v_{\theta c}}{t_c} \sim \nu \frac{v_{\theta c}}{r_c^2} \implies t_c \sim \frac{R^2}{\nu}$$

the time for viscous diffusion to cross the cavity. Define

$$V = \frac{v_\theta}{\Omega R}, \qquad \eta = \frac{r}{R}, \qquad \tau = \frac{t}{R^2/\nu}$$

so $\partial/\partial t = (\nu/R^2)\,\partial/\partial\tau$ and $\partial/\partial r = R^{-1}\partial/\partial\eta$ (chain rule). Term by term:

$$\frac{\partial v_\theta}{\partial t} = \frac{\Omega R\,\nu}{R^2}\frac{\partial V}{\partial \tau}, \qquad \nu\frac{\partial}{\partial r}\left[\frac{1}{r}\frac{\partial (r v_\theta)}{\partial r}\right] = \frac{\nu\,\Omega R}{R^2}\frac{\partial}{\partial \eta}\left[\frac{1}{\eta}\frac{\partial(\eta V)}{\partial \eta}\right]$$

Both sides carry the same factor $\nu\Omega/R$, which cancels:

$$\frac{\partial V}{\partial \tau} = \frac{\partial}{\partial \eta}\left[\frac{1}{\eta}\frac{\partial(\eta V)}{\partial \eta}\right]; \qquad \tau = 0:\ V = 0; \qquad \tau > 0:\ V(1,\tau) = 1,\ V(0,\tau) = 0$$

A problem without parameters: $V(\eta, \tau)$ is a universal function.

Temperature. With $\Theta = (T - T_0)/T_0$, each term of the energy equation becomes

$$\rho c\, T_0 \frac{\nu}{R^2}\frac{\partial \Theta}{\partial \tau} = \frac{k T_0}{R^2}\frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta \frac{\partial \Theta}{\partial \eta}\right) + \mu \Omega^2 \left[\eta \frac{\partial}{\partial \eta}\left(\frac{V}{\eta}\right)\right]^2$$

(the strain rate is $r\,\partial(v_\theta/r)/\partial r = \Omega\,\eta\,\partial(V/\eta)/\partial\eta$). Dividing by $\rho c T_0 \nu/R^2 = \mu c T_0/R^2$:

$$\frac{\partial \Theta}{\partial \tau} = \frac{1}{\mathrm{Pr}}\frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta \frac{\partial \Theta}{\partial \eta}\right) + \frac{\Omega^2 R^2}{c\,T_0}\left[\eta \frac{\partial}{\partial \eta}\left(\frac{V}{\eta}\right)\right]^2, \qquad \mathrm{Pr} = \frac{\mu c}{k}$$

with $\Theta = 0$ at $\tau = 0$, $\Theta = 0$ at $\eta = 1$ and $\partial\Theta/\partial\eta = 0$ at $\eta = 0$. The temperature depends on two parameters: the Prandtl number and $\Omega^2 R^2/(cT_0)$, the ratio of the kinetic energy of the wall to the thermal energy (an Eckert-type number; both are **extensions**, not defined in Chapter 6).

Discrepancy with the official solution

The key writes the conduction term as $\mathrm{Pr}\,\frac{1}{\eta}\frac{\partial}{\partial\eta}(\dots)$ with $\mathrm{Pr}=\mu c/k$; dividing kT_0/R^2 by $\mu c T_0/R^2$ gives $k/(\mu c)=1/\mathrm{Pr}$, as written above.

Torque from the Nondimensional Solution

The torque per unit length is the shear stress on the wall times the lever arm R times the perimeter $2\pi R$:

$$M = \tau_{r\theta}\big|_{r=R}\, R\, 2\pi R, \qquad \tau_{r\theta} = \mu\left[r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial\theta}\right] = \mu\Omega\,\eta\frac{\partial}{\partial\eta}\left(\frac{V}{\eta}\right)$$

($v_r=0$). Hence

$$M = 2\pi\mu\Omega R^2\left[\eta\frac{\partial}{\partial\eta}\left(\frac{V}{\eta}\right)\right]_{\eta=1} \quad\Longrightarrow\quad \frac{M}{2\pi\mu\Omega R^2} = f(\tau)$$

As $\tau\to\infty$, $V\to\eta$, so $V/\eta\to 1$ and its derivative vanishes: $f\to 0$ and $M\to 0$. Once the liquid turns as a rigid body there is no shear and no torque is needed to keep the case turning.

Comparison with the Π Theorem

The Π theorem gave $M/(\mu\Omega R^2)=f(\Pi_1,\Pi_2)$ with $\Pi_1=\Omega t$ and $\Pi_2=\rho\Omega R^2/\mu$. The nondimensional equations show that the two groups enter only through

$$\tau = \frac{t\nu}{R^2} = \frac{\Omega t}{\rho\Omega R^2/\mu} = \frac{\Pi_1}{\Pi_2}$$

so $M/(\mu\Omega R^2)=2\pi f(\Pi_1/\Pi_2)$: the two results are consistent, and the equations reduce the parametric dependence further than the Π theorem, which cannot see that ρ and t appear only in the combination $t\nu/R^2$ (Notes.pdf, p. 68: the nondimensional equations

give the minimum set of parameters).

Answers and Checks

- Π theorem: $M/(\mu\Omega R^2) = f(\Omega t, \rho\Omega R^2/\mu)$.
- Simplified problem: $v_r = 0$; $\rho v_\theta^2/r = \partial p/\partial r$; $\partial v_\theta/\partial t = \nu\,\partial_r[(1/r)\partial_r(r v_\theta)]$; $\rho c\,\partial T/\partial t = (k/r)\partial_r(r\partial_r T) + \mu[r\partial_r(v_\theta/r)]^2$, with the conditions of the table.
- $t \to \infty$: $v_\theta = \Omega r$, $p = p_1 + \frac{1}{2}\rho\Omega^2 R^2[(r/R)^2 - 1]$, $T = T_0$.
- Scales: $r_c = R$, $v_{\theta c} = \Omega R$, $t_c = R^2/\nu$; nondimensional velocity problem without parameters.
- Torque: $M/(2\pi\mu\Omega R^2) = f(\tau) = [\eta\,\partial_\eta(V/\eta)]_{\eta=1}$, which tends to 0 for $\tau \to \infty$.
- Dimensions: $\mu\Omega R^2$ is $ML^{-1}T^{-1}\cdot T^{-1}\cdot L^2 = MLT^{-2}$, a torque per unit length; $\Omega^2 R^2/(cT_0)$ is $L^2T^{-2}/(L^2T^{-2}\Theta^{-1}\cdot\Theta) = 1$.
- Pressure check: p is lowest on the axis, $p(0) = p_1 - \frac{1}{2}\rho\Omega^2 R^2$, as the centripetal acceleration requires.
- Check against the official key: DA14.pdf gives the same groups, equations, steady solution, scales, nondimensional problem and torque expression, with $\Pi_1' = \Pi_1/\Pi_2$; the coefficient of the conduction term is corrected above.